

Massive Transfusion Protocol

Policy Number	POL-CLN-043	Owner / Department	Blood Bank / Trauma Services
Effective Date	2024-03-04	Version	2.2
Last Revised	2026-04-09	Next Review	2028-04-09
Approved By	Patient Safety & Quality Committee	Applies To	Emergency department, operating rooms, obstetrics, ICU, blood bank, and trauma team staff

1. Purpose

To provide rapid, balanced delivery of blood components for patients with life-threatening hemorrhage, minimizing the coagulopathy of trauma and shock. This protocol reflects current AABB and massive-transfusion literature favoring a balanced 1:1:1 ratio of plasma, platelets, and red cells.

2. Scope

Applies to adult patients with, or at high risk of, massive hemorrhage — defined as transfusion of ≥ 10 units of red cells in 24 hours, ≥ 4 units in 1 hour with ongoing bleeding, or an Assessment of Blood Consumption (ABC) score ≥ 2 . Pediatric and obstetric variants are addressed in unit-specific appendices.

3. Activation

- Any physician or the trauma team leader may activate the MTP by calling the Blood Bank (dial 44) and stating “Activate MTP” with the patient location and two identifiers.
- Send a type-and-crossmatch sample immediately if not already on file; emergency-release group O red cells are issued until the sample resolves.
- Assign a runner to shuttle coolers between the bedside and the Blood Bank.
- Draw a baseline CBC, coagulation panel, fibrinogen, ionized calcium, and arterial blood gas.

IMPORTANT — Give tranexamic acid (TXA) 1 g IV over 10 minutes as early as possible — ideally within 3 hours of injury — followed by 1 g over 8 hours. Benefit falls sharply with delay, and TXA given beyond 3 hours after injury may increase mortality.

4. Component Ratios and Pack Contents

The Blood Bank issues components in pre-defined packs to maintain an approximately 1:1:1 ratio of plasma to platelets to red cells. Transfuse through a warmer with a rapid infuser; monitor for citrate-induced hypocalcemia.

MTP pack	Red cells	Plasma	Platelets
Pack 1	6 units	6 units	1 apheresis dose
Pack 2	6 units	6 units	1 apheresis dose

MTP pack	Red cells	Plasma	Platelets
Adjuncts	Cryoprecipitate if fibrinogen < 150 mg/dL	Calcium for ionized Ca^{2+} < 1.1 mmol/L	TXA per above

5. Targets and Termination

- Target fibrinogen ≥ 150 mg/dL, platelets $\geq 50 \times 10^9$ /L, INR < 1.5, ionized calcium ≥ 1.1 mmol/L, temperature $\geq 36^\circ\text{C}$, and pH ≥ 7.2 .
- Reassess with repeat labs after every pack or every 30–60 minutes.
- The treating physician deactivates the MTP by notifying the Blood Bank once hemostasis is achieved; return unused components promptly.

6. Severity & Reporting

Report MTP events and transfusion reactions per POL-RM-013; transfusion reactions also follow the facility hemovigilance process.

Severity	Example	Notify / timeframe
SEV-1	ABO-incompatible transfusion or MTP delay/failure contributing to death or permanent harm (sentinel event).	Attending, Administrator-on-Call, and Risk Management within 1 hour.
SEV-2	Acute transfusion reaction or component-delivery delay with major temporary harm.	Blood Bank medical director and Risk Management within 4 hours.
SEV-3	Protocol deviation (e.g. ratio error) with no patient harm.	Blood Bank supervisor within 24 hours via event report.
Near Miss	Mislabeled sample or wrong unit intercepted before transfusion.	Event report within 24 hours.

7. References

- AABB (Association for the Advancement of Blood & Biotherapies) standards and massive-transfusion guidance.
- Balanced-resuscitation (1:1:1) trial literature and CRASH-2 (tranexamic acid) evidence.
- Related: POL-CLN-042, POL-RM-013.